Superstore Profit Defense: Short-term Leakage Mitigation via SQL Audit and Long-term Retention Optimization via XGBoost

Phase 1: SQL Audit and Diagnostic Insights

1. The Macro-Level Landscape: Scaling the Footprint vs. Efficiency Dip

The initial audit establishes a baseline of Superstore's performance between 2014 and 2017. While the organization is in a rapid scaling phase, profitability is struggling to keep pace with volume expansion.

Revenue Throughput: The organization generated $2.29 million in total revenue.

Profitability Baseline: A total net profit of $284.96K was realized, resulting in an aggregate profit margin of 12.43%.

The Growth Paradox: Annual revenue surged from $483.14K to $733.42K, representing a 51.80% volume increase. However, the net margin dropped from its 2016 peak to 12.59% in 2017. This divergence suggests that recent growth is being bought at the cost of operational discipline.

2. The Surface Mirage: Preliminary Business Health Check

A high-level overview of performance metrics identifies several immediate red flags that appear to be primary drivers of the recent profit decline.

Regional Outliers: While the West and East regions maintain healthy double-digit margins, the Central region consistently underperforms, showing the lowest

average profitability in the entire portfolio.

The Furniture Burden: When breaking down profitability by category, Furniture emerges as a conspicuous burden. High logistical costs and aggressive discounting often result in near-zero or negative margins for large-scale items.

Logistics Inefficiency: An analysis of Ship Mode ROI reveals that First Class shipping is the most expensive mode. Preliminary data shows a high anomaly ratio where premium logistics are being utilized for orders that are already fundamentally unprofitable.

3. The Hidden Paradox: Peeling the Onion of Averages

Moving beyond surface averages, behavioral segmentation and RFM modeling uncover critical paradoxes that redefine strategic priorities.

The Leaky Bucket Paradox in the South: The South region delivers an industry-leading initial profit per capita, nearly double that of other regions. However, it suffers from the lowest 30-day retention rate at 9.07%. The South operates a harvesting model rather than a nurturing one, wasting expensive customer acquisition on one-off transactions.

The Bifurcation Paradox in the Central Region: The Central region is not broken; it is unfiltered. While averages are dragged down by a segment of Profit Drains (high-volume, low-margin deals costing $7,849), the region is also home to a powerhouse group of Champions contributing $19,007 in net profit.

Category Stickiness: Technology products boast a retention rate more than

double that of any other category. Customers entering the brand through a Technology purchase have a significantly higher Lifetime Value compared to price-sensitive Office Supplies customers.

4. **The Surgical Strike: Operational and Supply Chain Evidence**

Granular transactional auditing pinpoints the exact bleeding points of the organization.

The Toxic Discount Trap: Binders represent the single largest source of net loss within the Office Supplies category. Driven by a staggering average discount of 64.82%, these items attract low-value customers through unsustainable pricing.

The Machine Anomaly: High-ticket items like Machines show extreme volatility. Single transactions in specific states like Colorado have resulted in massive net losses due to excessive discount rates.

Geographic Hotspots of Loss: Profit leakage is concentrated in specific states where local management has abandoned margin protection. Central states like Texas and Illinois stand out as primary offenders.

Supply Chain Misallocation: There is a total lack of Service Level Priority. High-value Champions wait an average of 3.96 days for orders, while loss-making Profit Drains receive their orders in 3.86 days, essentially subsidizing fastest delivery for the least valuable transactions.

Failure of the Loss Leader Strategy: Market Basket Analysis debunks the myth that Binders pull customers into buying high-margin electronics. The most

frequent pairing is Binders plus Paper, representing low-value bundling rather than effective cross-selling.

5. **Audit Conclusion: Where the Blood is Leaking**

   The SQL audit confirms that profit leakage is not a result of market conditions, but a systemic failure in pricing discipline and resource allocation.

   Specific sub-categories (Binders, Machines) are being exploited by cherry-picking customers.

   Premium logistics (First Class) are being democratized and wasted on unprofitable transactions.

   High-acquisition regions like the South are failing to plug the leaky bucket of customer churn.

   Aggressive discounting is creating a race to the bottom rather than a path to long-term profitability.

These findings provide the necessary features and business logic to transition into the next phase: predicting long-term customer retention via Machine Learning.

Phase 2: Machine Learning Prediction and Deep Insights

1. **Transition from Diagnostic to Proactive Strategy**

   While the SQL audit successfully identified where the profit leakage occurred in the past, the Machine Learning phase aims to answer the critical question: Can the bleeding be stopped before it starts? By building a predictive engine, the

organization can identify high-risk customers at the point of their first transaction, allowing for immediate intervention in pricing and logistics allocation.

2. **Model Architecture and the 180-Day Retention Window**

To capture the true purchase cycle of office supplies and furniture, the retention window was extended from a 30-day tactical view to a 180-day strategic window. This adjustment ensures that the model captures long-term loyalty rather than just seasonal noise.

Target Variable: A binary label (is_retained) representing whether a customer placed a subsequent order within 180 days of their initial purchase.

Algorithm Choice: XGBoost Classifier was selected for its superior ability to handle non-linear relationships and its built-in capacity to manage imbalanced datasets.

Handling Class Imbalance: Given that the majority of customers do not return within the specified window, the model utilized a scale_pos_weight strategy to penalize the misclassification of the minority (retained) class.

3. **Performance Metrics: Moving Beyond the Accuracy Trap**

In a highly imbalanced environment, simple Accuracy is a deceptive metric. A model that predicts everyone will churn would still be 80% accurate but commercially useless.

F1 Score: The model achieved an F1 Score of 0.4562. This represents a significant improvement over the baseline, proving the model possesses genuine

predictive power in identifying future repeat customers.

ROC-AUC: The average ROC-AUC reached 0.5292, indicating that the model is 2.92% better than random chance at distinguishing between a one-off buyer and a long-term Loyalist.

4. **Feature Importance: AI-Driven Validation of Structural Risks**

The refined XGBoost model, optimized for class imbalance, identifies a clear hierarchy of predictors that independently validate the initial SQL audit findings. The top features reveal that retention is not random, but deeply rooted in geography and service expectations.

Region South as the Primary Predictor: Region South emerged as the most significant feature in predicting whether a customer returns. This confirms the SQL finding that the South is a strategic anomaly. It is winning the first sale but structurally losing the war on customer loyalty, regardless of sales volume or category.

Logistics as a Behavioral Proxy: Ship Mode dominates the top five features, with First Class and Standard Class showing high predictive importance. This suggests that how a customer chooses to receive their goods is a powerful indicator of their persona. Premium logistics users (First Class) and steady procurement users (Standard Class) represent distinct behavioral cohorts with vastly different retention probabilities.

Discounting as an Overestimated Factor: While Discounts remain in the top 10, their importance is far less than that of Region and Ship Mode. Offering discount coupons (especially for Office Supplies) not only fails to effectively improve customer loyalty, but also attracts customers who are simply looking to save money.

5. **Predictive Conclusion: The Early Warning System**

The ML phase proves that the bleeding can indeed be stopped. By analyzing the characteristics of a customer's first order, we can predict their 180-day retention probability with 52.92% confidence. This allows the organization to move away from a one-size-fits-all approach and instead deploy resources based on predicted Lifetime Value (LTV).

The model confirms that the Profit Drains identified in the SQL audit are not just unprofitable in the short term, but are fundamentally unlikely to ever become profitable over time. This insight sets the stage for the final phase: the Surgical Recovery Plan.

Phase 3: Strategic Recommendations – The Surgical Recovery Plan

Based on the synthesis of historical audit data and predictive modeling, we propose a targeted recovery plan centered on the Retention-Profitability Strategy Matrix. This plan shifts the organization from volume-at-any-cost to value-driven acquisition.

1. **The Strategic Matrix Framework**

The 2x2 Matrix serves as the operational compass for resource allocation, categorizing every new customer at the point of their first transaction based on two dimensions: First-Order Profit Margin (SQL Insight) and Predicted Retention Probability (ML Insight).

2. **Segment-Specific Interventions**

The Champions (High Profit, High Retention)

Strategy: Protect and Reward.

Action: These customers do not require discounts to remain loyal. Re-allocate saved discount capital into premium service. Establish a Tiered Service Level Agreement (SLA) where First Class and Same Day shipping are exclusively reserved for this segment to reinforce loyalty.

The Leaky Bucket (High Profit, Low Retention)

Strategy: Retention Focus.

Action: This segment, prevalent in the South region, wins the first sale but loses the customer. Implement an automated Post-Purchase Engagement Program. Within 30 days of the first order, offer a Targeted Loyalty Credit specifically for Technology products to anchor the customer into a high-stickiness category.

Strategic Growth (Low Profit, High Retention)

Strategy: Upsell and Cross-sell.

Action: These are loyal customers currently buying low-margin commodities. Utilize Market Basket Analysis (MBA) to break the Binders-to-Paper cycle.

Implement a recommendation engine that cross-sells high-margin Technology accessories (Phones, Accessories) to improve the Profit Margin of their subsequent orders.

The Vampires (Low Profit, Low Retention)

Strategy: Cost Control and Mitigation.

Action: These customers, often found in the Central region (Texas/Illinois), represent a systemic drain on resources. Implement a Hard Cap on Discounts for the Office Supplies category. For any order predicted with a low retention probability, the system must default to Standard Class/Ground shipping to eliminate the logistics subsidy for one-off buyers.

3. **Regional Management Re-alignment**

South Region: Shift KPI focus from Gross Sales to 30-day & 180-day Retention Rate and LTV Growth. The region must transition from a harvesting model to a nurturing model to plug the leaky bucket.

Central Region: Conduct an immediate pricing audit for the Texas and Illinois branches. Eliminate the discount arbitrage loophole that attracts cherry-picking customers who offer zero long-term value.

4. **Operational Pivot: Logistics as a Retention Tool**

Stop the Democratization of Speed: The audit proved that loss-making Profit Drains currently receive their orders as fast as Champions. Logistics must be utilized as a tool for retention, not a baseline subsidy. High-speed tiers should be

dynamically unlocked only for customers with high predicted retention scores or healthy profit margins.

5. **Final Conclusion: Scaling with Discipline**

By integrating SQL-driven diagnostic auditing with XGBoost predictive analytics, Superstore can stop the immediate profit leakage while building a sustainable engine for growth. The goal is no longer just to sell, but to acquire the right customers at the right price with the right service level. This data-driven roadmap ensures that as the organization scales, its efficiency and profitability scale with it.